

Supplementary Appendix

This appendix has been provided by the authors to give readers additional information about their work.

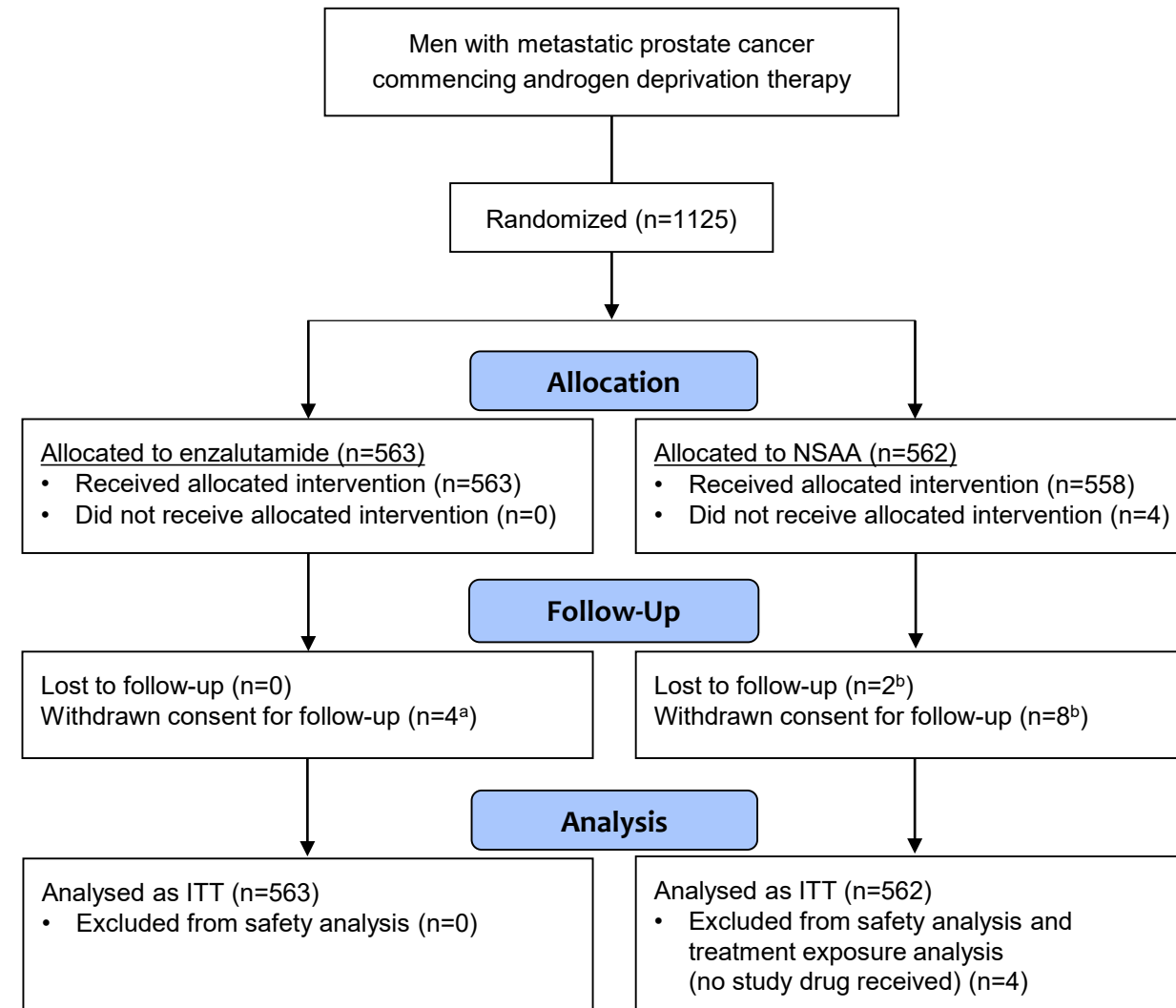
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ENZAMET Trial: Supplementary Appendix

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Supplementary Figure S1. CONSORT diagram.

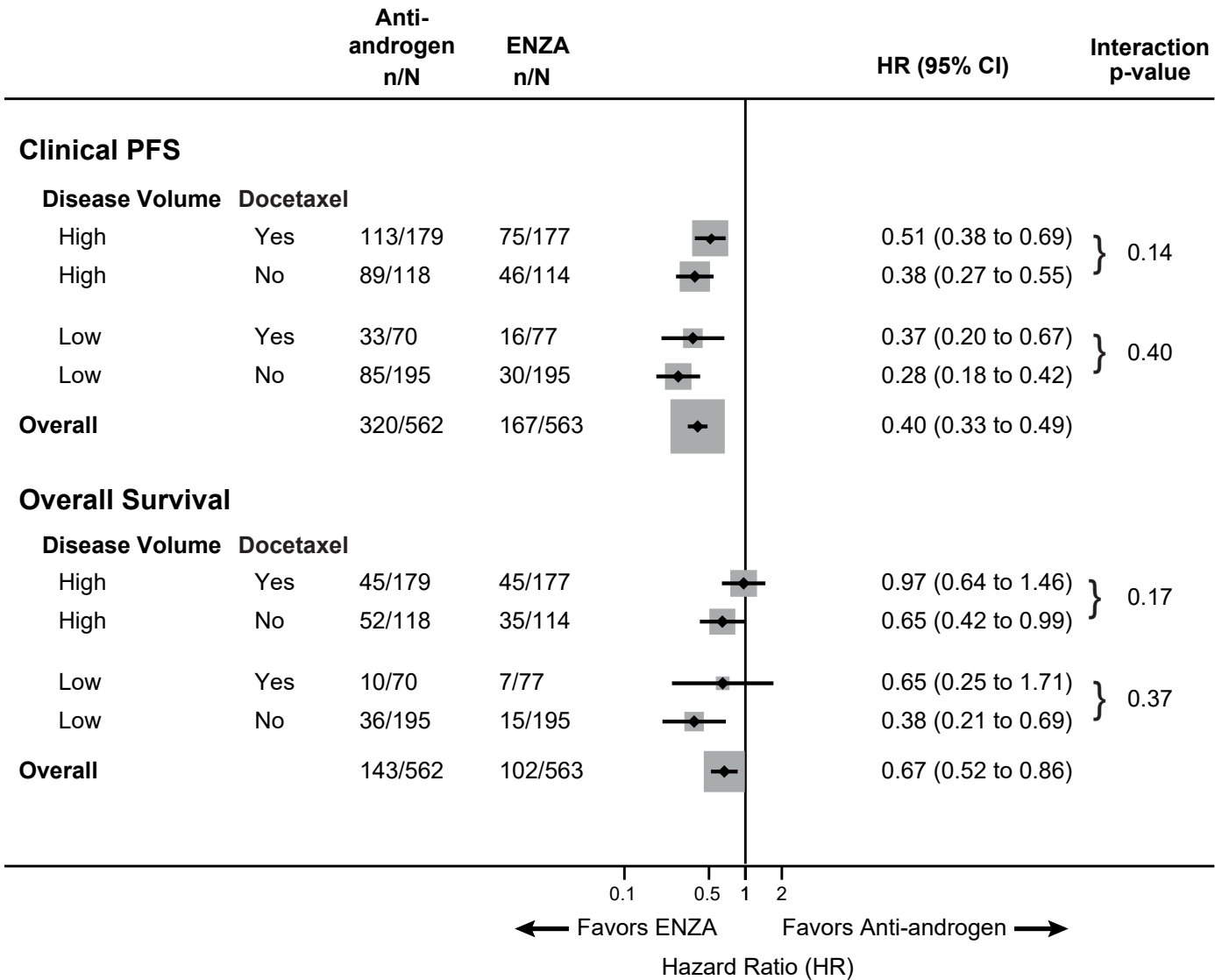


^a Median survival follow-up for the 4 enzalutamide patients who withdrew consent was 7.2 months (IQR: 2.0 to 24.4).

^b Median survival follow-up for the 10 NSAA patients who withdrew consent or were lost to follow-up was 7.7 months (IQR: 0.4 to 12.0). Two of these patients withdrew consent immediately post-randomization.

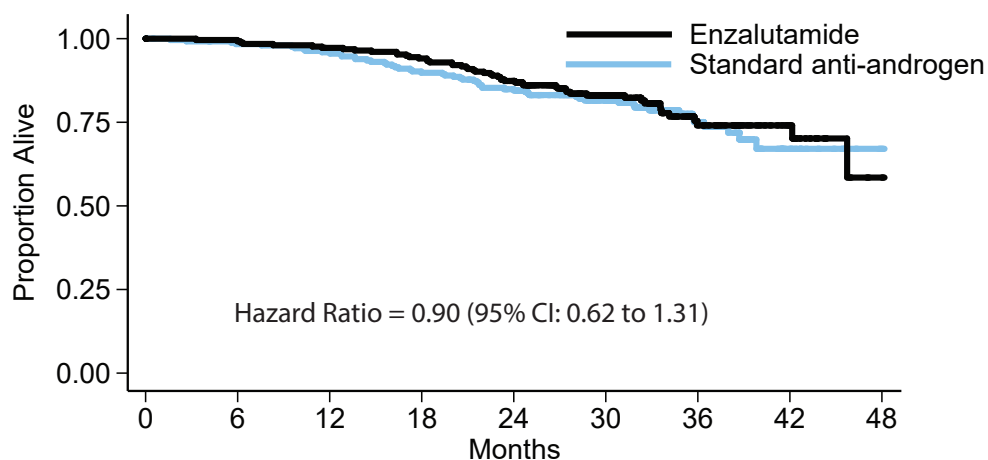
Supplementary Figure S2: Forest plots for disease burden and use of docetaxel

Forest plots for clinical Progression-Free Survival and Overall Survival, for anti-androgen and enzalutamide groups according to disease burden (high volume or low volume) and use of docetaxel. Abbreviations: CI, confidence interval; cPFS, clinical Progression-Free Survival; ENZA, enzalutamide; HR, hazard ratio.



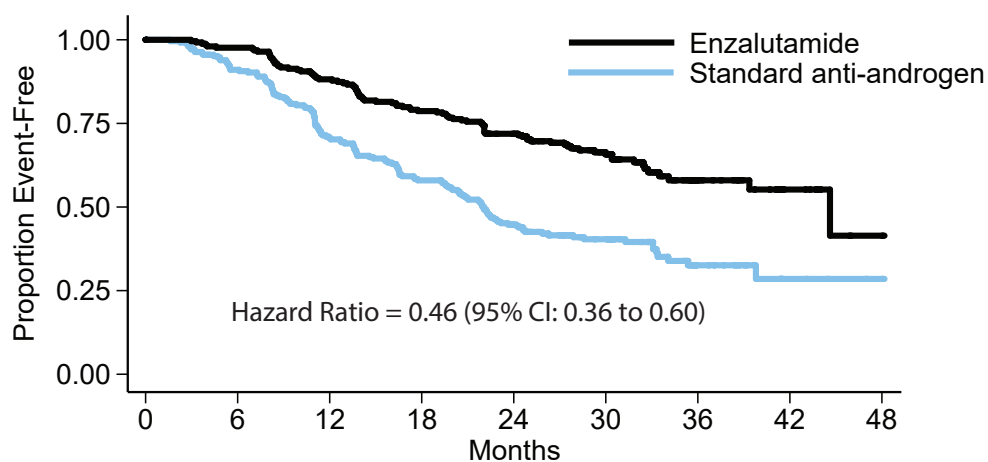
Supplementary Figure S3: Kaplan-Meier curves for docetaxel subgroup

(a) Overall Survival



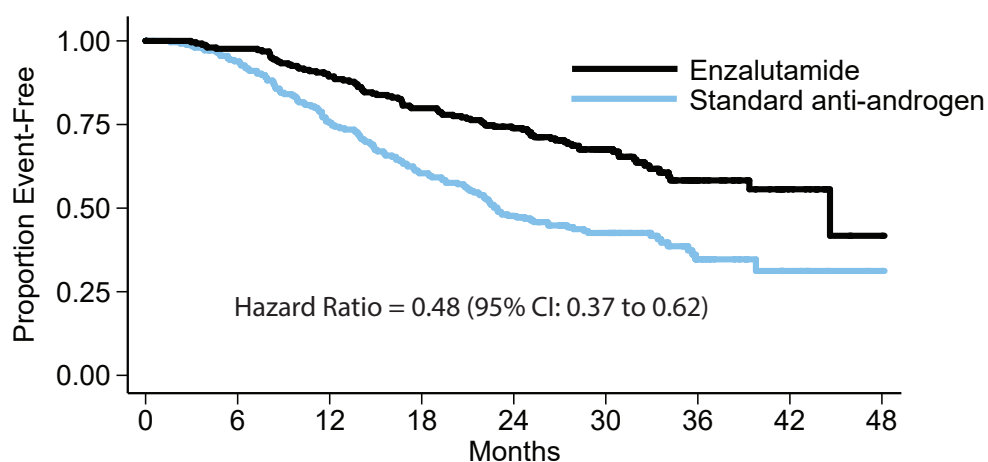
Number at risk									
Standard anti-androgen	249	241	235	220	203	135	56	13	2
Enzalutamide	254	252	246	238	210	139	54	19	3

(b) PSA Progression-Free Survival



Number at risk									
Standard anti-androgen	249	223	173	142	105	67	17	4	1
Enzalutamide	254	248	223	198	171	101	31	10	2

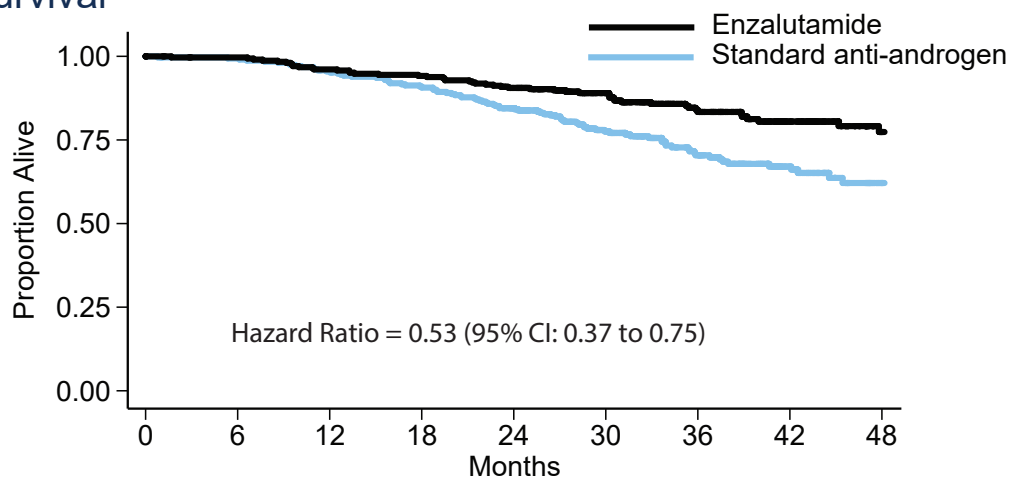
(c) Clinical Progression-Free Survival



Number at risk									
Standard anti-androgen	249	230	185	148	112	73	21	6	1
Enzalutamide	254	248	226	202	178	109	35	12	2

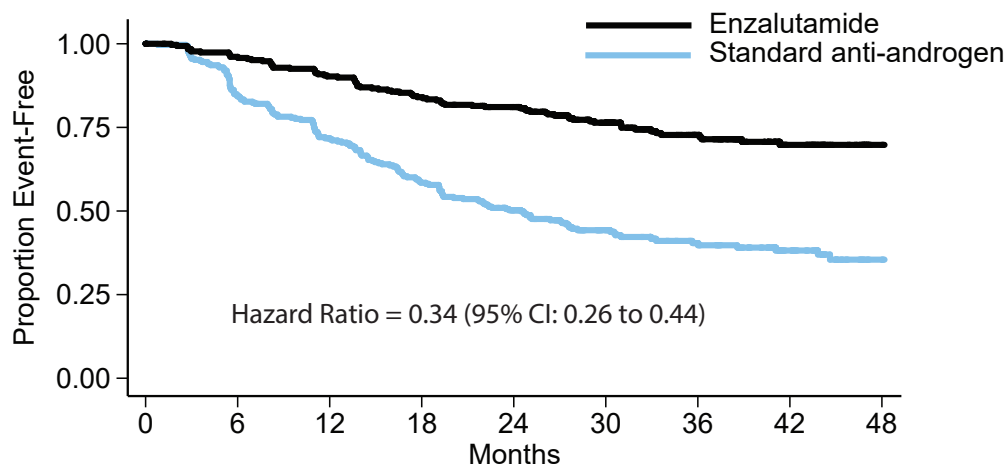
Supplementary Figure S4: Kaplan-Meier curves for no-docetaxel subgroup

(a) Overall Survival



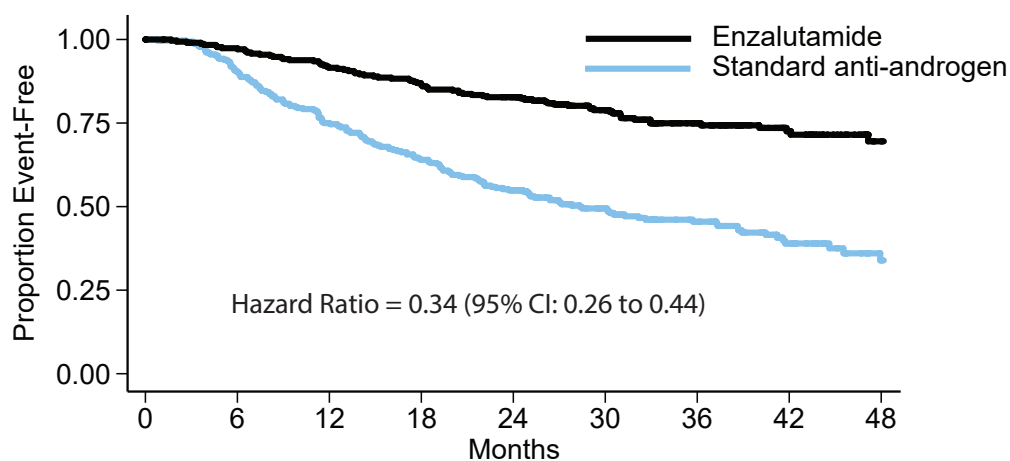
Number at risk									
Standard anti-androgen	313	310	296	281	249	176	118	73	30
Enzalutamide	309	306	295	289	270	201	135	87	42

(b) PSA Progression-Free Survival



Number at risk									
Standard anti-androgen	313	263	222	180	144	94	61	40	16
Enzalutamide	309	295	277	257	240	168	115	67	32

(c) Clinical Progression-Free Survival



Number at risk									
Standard anti-androgen	313	282	233	198	160	109	75	44	16
Enzalutamide	309	299	281	266	246	175	121	72	34

Supplementary Table S1: ENZAMET sites, investigators, key sponsor and coordinating center staff

ENZAMET sites and investigators:

Australia:

Ehtesham Abdi, The Tweed hospital; Suzanne Allan, Gold Coast Hospital; Patricia Bastick, St George Hospital; Stephen Begbie, Port Macquarie Base Hospital, North Coast Cancer Institute; Robert Blum, Bendigo Hospital; Karen Briscoe, Coffs Harbour Health Campus; Daniel Brungs, Wollongong Hospital; Sean Bydder, Sir Charles Gardiner Hospital; Bala Renuka Chittajallu, Riverina Cancer Centre; Michelle Cronk, Nambour General hospital; Katharine Cuff, Princess Alexandra Hospital Brisbane; Ian Davis, Eastern Health - Box Hill Hospital; Anthony Dowling, St Vincent's Hospital Melbourne; Mark Frydenberg, Australian Urology Associates; Matthew George, Tamworth Hospital; Lisa Horvath, Chris O'Brien Lifehouse; Elizabeth Hovey, Prince of Wales Hospital; Anthony Joshua, St Vincent's Hospital Sydney; Narayan Karanth, Royal Darwin Hospital; Ganessan Kichenadasse, Flinders Medical Centre and RGH; Laurence Krieger, Northern Cancer Institute; Gavin Marx, Sydney Adventist Hospital; Maitham Mathlum, Geelong Hospital; Louise Nott, Royal Hobart Hospital; Zulfiquer Otty, Townsville Hospital; Francis Parnis, Ashford Cancer Centre; David Pook, Monash Cancer Centre - Moorabbin; Shahneen Sandhu, Peter MacCallum Cancer Centre; Sanjeev Sewak, Peninsula South Eastern Haematology & Oncology Group; Amanda Stevanovic, Nepean Hospital; Martin Stockler, Concord Repatriation General Hospital; Aneta Suder, Royal Brisbane and Women's Hospital; Hsiang Tan, Royal Adelaide Hospital; Javier Torres, Goulburn Valley Health; Simon Troon, Fiona Stanley Hospital; Craig Underhill, Border Medical Oncology; Andrew Weickhardt, Austin Hospital; Robert Zielinski, Orange Health Service.

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Ireland:

John McCaffrey, Mater Private Hospital, and Mater Misericordiae University Hospital; Ray McDermott, Adelaide and Meath Hospital – National Children Hospital, and St Vincent's University Hospital; Patrick Morris, Beaumont Hospital; Miriam O'Connor, University Hospital Waterford; Paul Donnellan, Galway University Hospital; Dearbhaile O'Donnell, St James Hospital.

New Zealand:

Jim Edwards, Christchurch Hospital; Peter Fong, Auckland City Hospital; Alvin Tan, Waikato General Hospital.

United Kingdom:

Simon Chowdhury, Guy's and St. Thomas Hospital; Simon Crabb, University Hospital Southampton; Omar Khan, Great Western General Hospital); Vincent Khoo, The Royal Marsden NHS Foundation Trust; Graham Macdonald, Aberdeen Royal Infirmary; Heather Payne, University of London Hospital; Angus Robinson, BSUH (Royal Sussex Hospital); Jonathon Shamash, Barts Health NHS Trust, Bartholomew's Hospital; John Staffurth, Velindre Cancer Centre; Carys Thomas, Kent & Canterbury Hospital; Alastair Thomson, Royal Cornwall Hospital.

USA:

Christopher Sweeney, Dana-Farber Cancer Institute

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ANZUP Independent Data & Safety Monitoring Committee: Ian Olver, Richard Bell, Peta Forder

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Supplementary Table S2: Participant baseline characteristics

* Stage at first diagnosis. ** Commenced within 12 weeks of randomization. Abbreviations: ADT, androgen deprivation therapy; BMI, body mass index; IQR, interquartile range; LHRHA, luteinizing hormone releasing hormone agonist / antagonist; SD, standard deviation.

Characteristic		Anti-androgen (N=562)	Enzalutamide (N=563)
Age (years)	Mean (SD)	68.8 (8.3)	68.9 (8.1)
	Median (IQR)	69.0 (63.6 to 74.5)	69.2 (63.2 to 74.5)
ECOG Performance Status	0	405 (72.1%)	405 (71.9%)
	1	151 (26.9%)	150 (26.6%)
	2	6 (1.1%)	8 (1.4%)
Country	Australia	321 (57.1%)	324 (57.5%)
	Canada	107 (19.0%)	97 (17.2%)
	Ireland	43 (7.7%)	39 (6.9%)
	New Zealand	19 (3.4%)	20 (3.6%)
	UK	50 (8.9%)	63 (11.2%)
	United States	22 (3.9%)	20 (3.6%)
Planned use of early docetaxel	No	313 (55.7%)	309 (54.9%)
	Yes	249 (44.3%)	254 (45.1%)
Volume of disease	High	297 (52.8%)	291 (51.7%)
	Low	265 (47.2%)	272 (48.3%)
Prior local therapy		235 (41.8%)	238 (42.3%)
Bone anti-resorptive therapy		58 (10.3%)	55 (9.8%)
ACE-27 stratum	0-1	419 (74.6%)	422 (75.0%)
	2-3	143 (25.4%)	141 (25.0%)
Number of bone metastases	1 - 3	165 (29.4%)	161 (28.6%)
	4 or more	294 (52.3%)	288 (51.2%)
	None	103 (18.3%)	114 (20.2%)
Visceral metastases	No	495 (88.1%)	501 (89.0%)
	Yes	67 (11.9%)	62 (11.0%)
	Adrenal	6 (1.1%)	1 (0.2%)
	Liver	11 (2.0%)	13 (2.3%)
	Lung	48 (8.5%)	55 (9.8%)
	Pleura	5 (0.9%)	3 (0.5%)
	Other	4 (0.7%)	0 (0.0%)
T stage*	T0	3 (0.5%)	1 (0.2%)
	T1	30 (5.3%)	23 (4.1%)
	T2	131 (23.3%)	124 (22.0%)
	T3	198 (35.2%)	229 (40.7%)
	T4	60 (10.7%)	56 (9.9%)
	TX	72 (12.8%)	54 (9.6%)
	Unknown	68 (12.1%)	76 (13.5%)

Characteristic		Anti-androgen (N=562)	Enzalutamide (N=563)
N stage*	N0	237 (42.2%)	226 (40.1%)
	N1	194 (34.5%)	205 (36.4%)
	NX	65 (11.6%)	63 (11.2%)
	Unknown	66 (11.7%)	69 (12.3%)
M stage*	M0	157 (27.9%)	155 (27.5%)
	M1	347 (61.7%)	335 (59.5%)
	MX	27 (4.8%)	27 (4.8%)
	Unknown	31 (5.5%)	46 (8.2%)
Months since primary diagnosed	Mean (SD)	23.9 (40.2)	26.9 (45.3)
	Median (IQR)	3.1 (1.7 to 32.7)	2.0 (2.0 to 39.1)
Months since metastases diagnosed	Mean (SD)	3.1 (7.2)	2.9 (6.9)
	Median (IQR)	1.9 (1.0 to 2.8)	1.9 (0.9 to 2.8)
Gleason Score	≤7	163 (29.0%)	152 (27.0%)
	8-10	321 (57.1%)	335 (59.5%)
	Missing	78 (13.9%)	76 (13.5%)
BMI	Mean (SD)	28.1 (4.8)	28.5 (5.0)
	Median (IQR)	27.7 (25.0 to 30.7)	27.8 (25.2 to 31.1)
Prior adjuvant ADT		40 (7.1%)	58 (10.3%)
Prior anti-androgen**		316 (56.2%)	285 (50.6%)
Prior LHRHA**		418 (74.4%)	411 (73.0%)
Bilateral orchiectomy		8 (1.4%)	5 (0.9%)
Prior docetaxel**		83 (14.8%)	95 (16.9%)

Supplementary Table S3. Survival at 36 months by subgroups

Subgroup	Level	Proportion Alive at 36 Months (95% CI)	
		Anti-androgen	Enzalutamide
Volume of disease	Low	0.82 (0.75 to 0.87)	0.90 (0.84 to 0.93)
	High	0.64 (0.58 to 0.70)	0.71 (0.64 to 0.76)
Early Docetaxel Planned	Yes	0.75 (0.68 to 0.81)	0.74 (0.66 to 0.80)
	No	0.70 (0.64 to 0.76)	0.83 (0.78 to 0.87)
ACE-27 Co-morbidity Score	2-3	0.66 (0.56 to 0.75)	0.74 (0.65 to 0.82)
	0-1	0.75 (0.70 to 0.79)	0.82 (0.77 to 0.85)
Anti-resorptive therapy	Yes	0.77 (0.61 to 0.87)	0.69 (0.52 to 0.81)
	No	0.72 (0.67 to 0.76)	0.81 (0.76 to 0.85)
Region	Ireland/UK	0.72 (0.59 to 0.81)	0.73 (0.60 to 0.82)
	N America	0.72 (0.60 to 0.80)	0.77 (0.64 to 0.86)
	ANZ	0.73 (0.67 to 0.78)	0.82 (0.77 to 0.86)
Gleason score	≤7	0.88 (0.81 to 0.92)	0.92 (0.85 to 0.95)
	8-10	0.70 (0.64 to 0.76)	0.77 (0.72 to 0.82)
ECOG Performance Status	1-2	0.55 (0.45 to 0.64)	0.69 (0.59 to 0.76)
	0	0.79 (0.74 to 0.83)	0.84 (0.79 to 0.88)
Age (Years)	≥70	0.67 (0.60 to 0.73)	0.80 (0.73 to 0.85)
	<70	0.77 (0.71 to 0.82)	0.79 (0.73 to 0.84)
Visceral metastases	Yes	0.73 (0.59 to 0.82)	0.70 (0.55 to 0.81)
	No	0.72 (0.67 to 0.77)	0.81 (0.76 to 0.85)
Prior Local Treatment	Yes	0.76 (0.68 to 0.82)	0.83 (0.77 to 0.88)
	No	0.70 (0.64 to 0.75)	0.77 (0.71 to 0.82)
Overall	All	0.72 (0.68 to 0.76)	0.80 (0.75 to 0.83)

Abbreviations: ANZ, Australia and New Zealand; CI, confidence intervals; ECOG, Eastern Cooperative Oncology Group; N America, North America (USA and Canada); UK, United Kingdom.

Supplementary Table S4: Post-study therapies

Participants with Clinical Progression-Free Survival Endpoint		
Therapy	Anti-androgen (N=320)	Enzalutamide (N=167)
Enzalutamide	141 (44.1%)	0 (0%)
Abiraterone acetate*	113 (35.3%)	46 (27.5%)
Other novel antiandrogen	2 (0.6%)	1 (0.6%)
Docetaxel	69 (21.6%)	45 (26.9%)
Cabazitaxel	64 (20.0%)	34 (20.4%)
Other chemotherapy	20 (6.3%)	22 (13.2%)
Immune checkpoint inhibitor	6 (1.9%)	10 (6.0%)
Lutetium-177 PSMA	4 (1.3%)	6 (3.6%)
PARP inhibitor	11 (3.4%)	4 (2.4%)
Radium-223 dichloride	22 (6.9%)	14 (8.4%)
Sipuleucel-T	2 (0.6%)	0 (0%)
No treatment yet	49 (15.3%)	55 (32.9%)

Participants who Died of Prostate Cancer		
Therapy	Anti-androgen (N=115)	Enzalutamide (N=88)
Enzalutamide	52 (45.2%)	0 (0%)
Abiraterone acetate*	41 (35.7%)	27 (30.7%)
Other novel antiandrogen	1 (0.9%)	1 (1.1%)
Docetaxel	40 (34.8%)	29 (33.0%)
Cabazitaxel	35 (30.4%)	19 (21.6%)
Other chemotherapy	13 (11.3%)	14 (15.9%)
Immune checkpoint inhibitor	1 (0.9%)	5 (5.7%)
Lutetium-177 PSMA	1 (0.9%)	3 (3.4%)
PARP inhibitor	4 (3.5%)	1 (1.1%)
Radium-223 dichloride	7 (6.1%)	9 (10.2%)
Sipuleucel-T	0 (0%)	0 (0%)
No treatment	13 (11.3%)	28 (31.8%)

Note: These treatments were given after a progression event in all four cases where patient or physician preference was listed as the reason.

* Abiraterone acetate was combined with prednisone or prednisolone.

Abbreviations: PARP, poly-ADP ribose polymerase; PSMA, prostate-specific membrane antigen ligand.

Supplementary Table S5: Treatment exposure and reasons for discontinuation

Characteristic		Anti-androgen (N=558)	Enzalutamide (N=563)
Proportion on treatment at 12m (95% CI)		0.72 (0.68 to 0.75)	0.86 (0.83 to 0.88)
Proportion on treatment at 24m (95% CI)		0.47 (0.43 to 0.52)	0.74 (0.70 to 0.77)
Proportion on treatment at 36m (95% CI)		0.34 (0.29 to 0.38)	0.62 (0.57 to 0.66)
Has patient ceased anti-androgen treatment?	Yes	356 (63.8%)	201 (35.7%)
Reason for permanently ceasing anti-androgen treatment*	Clinical Progression (Imaging)	144 (40.4%)	88 (43.8%)
	Clinical Progression (Symptoms)	55 (15.4%)	32 (15.9%)
	Clinical Progression (Anti-cancer treatment)	45 (12.6%)	7 (3.5%)
	Adverse event	14 (3.9%)	33 (16.4%)
	Clinician preference	58 (16.3%)	13 (6.5%)
	Death	7 (2.0%)	6 (3.0%)
	Other	6 (1.7%)	8 (4.0%)
	Patient preference	27 (7.6%)	14 (7.0%)

* Denominator is number of participants who have ceased treatment. Note: Investigators were allowed to select more than one criterion for clinical progression, however only one criterion is shown in the table with sequence for attribution being: (1) imaging, (2) symptoms, (3) anti-cancer treatment, (4) Other

Abbreviation: CI, confidence interval; m, months.

Supplementary Table S6: All Adverse Events by Worst Grade*

System Organ Class/CTCAE Term		Anti-androgen						Enzalutamide						TOTAL
		Worst grade						Worst grade						
		1	2	3	4	5	N	1	2	3	4	5	N	
Blood and lymphatic system disorders	Total**	75	26	29	9	_	139	90	27	34	7	_	158	297
	Anemia	60	16	4	1	_	81	64	21	4	_	_	89	170
	Blood and lymphatic system disorders - Other, specify	15	10	_	_	_	25	22	5	1	_	_	28	53
	Bone marrow hypocellular	_	_	_	_	_	_	2	_	_	_	_	2	2
	Febrile neutropenia	_	_	24	8	_	32	_	_	30	7	_	37	69
	Leukocytosis	_	_	1	_	_	1	_	_	_	_	_	_	1
	Lymph node pain	_	_	_	_	_	_	1	_	_	_	_	1	1
	Thrombotic thrombocytopenic purpura	_	_	_	_	_	_	1	_	_	_	_	1	1
Cardiac disorders	Total**	26	31	19	2	1	79	48	34	25	3	1	111	190
	Acute coronary syndrome	_	1	4	_	_	5	_	2	2	1	_	5	10
	Aortic valve disease	1	1	1	_	_	3	_	_	1	_	_	1	4
	Atrial fibrillation	_	6	3	_	_	9	2	6	4	_	_	12	21
	Atrial flutter	_	1	2	_	_	3	_	1	3	1	_	5	8
	Atrioventricular block complete	_	1	_	_	_	1	_	_	1	_	_	1	2
	Atrioventricular block first degree	_	_	_	_	_	_	_	1	_	_	_	1	1
	Cardiac arrest	_	_	_	_	1	1	_	_	_	_	_	_	1

System Organ Class/CTCAE Term	Anti-androgen						Enzalutamide						TOTAL
	Worst grade						Worst grade						
	1	2	3	4	5	N	1	2	3	4	5	N	
	12	6	_	_	_	18	15	11	1	1	_	28	46
Cardiac disorders - Other, specify													
Chest pain - cardiac	2	3	2	_	_	7	8	8	3	_	_	19	26
Heart failure	_	2	2	1	_	5	_	_	5	_	_	5	10
Left ventricular systolic dysfunction	_	_	1	_	_	1	_	_	_	_	_	_	1
Mitral valve disease	_	_	_	_	_	_	_	1	_	_	_	1	1
Mobitz (type) II atrioventricular block	_	_	_	_	_	_	1	_	_	_	_	1	1
Myocardial infarction	_	_	1	1	_	2	_	1	4	_	1	6	8
Palpitations	3	2	_	_	_	5	9	1	_	_	_	10	15
Paroxysmal atrial tachycardia	_	1	1	_	_	2	_	_	_	_	_	_	2
Pericardial effusion	_	1	_	_	_	1	_	_	_	_	_	_	1
Pericarditis	_	1	_	_	_	1	1	1	_	_	_	2	3
Sinus bradycardia	4	1	_	_	_	5	5	_	_	_	_	5	10
Sinus tachycardia	2	1	_	_	_	3	7	1	_	_	_	8	11
Supraventricular tachycardia	_	3	2	_	_	5	_	_	1	_	_	1	6
Ventricular arrhythmia	2	_	_	_	_	2	_	_	_	_	_	_	2
Ear and labyrinth disorders	16	4	2	_	_	22	33	15	3	_	_	51	73
Total**													
Ear and labyrinth disorders - Other, specify	6	1	_	_	_	7	9	4	_	_	_	13	20
Ear pain	1	_	_	_	_	1	2	1	_	_	_	3	4
Hearing impaired	_	_	1	_	_	1	5	2	2	_	_	9	10

System Organ Class/CTCAE Term		Anti-androgen						Enzalutamide						TOTAL
		Worst grade						Worst grade						
		1	2	3	4	5	N	1	2	3	4	5	N	
Endocrine disorders	Middle ear inflammation	—	1	—	—	—	1	—	—	—	—	—	—	1
	Tinnitus	5	1	—	—	—	6	10	5	—	—	—	15	21
	Vertigo	4	1	—	—	—	5	7	3	1	—	—	11	16
	Vestibular disorder	—	—	1	—	—	1	—	—	—	—	—	—	1
	Total**	16	6	—	—	—	22	24	6	—	—	—	30	52
	Cushingoid	2	—	—	—	—	2	—	—	—	—	—	—	2
	Endocrine disorders - Other, specify	13	5	—	—	—	18	23	2	—	—	—	25	43
	Hyperparathyroidism	—	—	—	—	—	—	—	1	—	—	—	1	1
	Hyperthyroidism	—	—	—	—	—	—	—	1	—	—	—	1	1
	Hypoparathyroidism	—	—	—	—	—	—	—	1	—	—	—	1	1
	Hypothyroidism	1	—	—	—	—	1	1	1	—	—	—	2	3
	Virilization	—	1	—	—	—	1	—	—	—	—	—	—	1
Eye disorders	Total**	55	19	9	1	—	84	120	26	12	1	—	159	243
Blurred vision	5	1	1	—	—	7	8	2	—	—	—	10	17	
Cataract	1	1	6	—	—	8	—	1	7	—	—	8	16	
Conjunctivitis	2	3	—	—	—	5	5	4	—	—	—	9	14	
Corneal ulcer	—	—	1	—	—	1	—	—	—	—	—	—	1	
Dry eye	11	—	—	—	—	11	14	2	—	—	—	16	27	
Eye disorders - Other, specify	16	6	1	1	—	24	30	6	2	—	—	38	62	

System Organ Class/CTCAE Term	Anti-androgen						Enzalutamide						TOTAL	
	Worst grade						Worst grade							
	1	2	3	4	5	N	1	2	3	4	5	N		
Gastrointestinal disorders	Eye pain	1	—	—	—	—	1	2	2	—	—	—	4	5
	Eyelid function disorder	—	—	—	—	—	—	—	1	—	—	—	1	1
	Floaters	—	—	—	—	—	—	1	—	—	—	—	1	1
	Glaucoma	1	1	—	—	—	2	1	1	—	1	—	3	5
	Keratitis	—	1	—	—	—	1	—	—	1	—	—	1	2
	Optic nerve disorder	—	1	—	—	—	1	—	—	—	—	—	—	1
	Photophobia	2	2	—	—	—	4	2	1	—	—	—	3	7
	Retinal detachment	—	—	—	—	—	—	1	—	1	—	—	2	2
	Vitreous hemorrhage	2	1	—	—	—	3	1	—	—	—	—	1	4
	Watering eyes	14	2	—	—	—	16	55	6	1	—	—	62	78
	Total**	429	121	27	—	1	578	545	167	19	1	—	732	1310
	Abdominal distension	1	—	—	—	—	1	2	2	—	—	—	4	5
	Abdominal pain	20	2	4	—	—	26	31	6	2	—	—	39	65
	Anal fistula	—	—	—	—	—	—	—	—	2	—	—	2	2
Anal hemorrhage	1	1	—	—	—	2	3	—	—	—	—	3	5	
Anal pain	2	1	—	—	—	3	3	—	—	—	—	3	6	
Ascites	—	1	—	—	—	1	—	—	—	—	—	—	1	
Bloating	2	1	—	—	—	3	7	4	—	—	—	11	14	
Colitis	—	—	2	—	—	2	—	1	1	—	—	2	4	

System Organ Class/CTCAE Term	Anti-androgen						Enzalutamide						TOTAL
	Worst grade						Worst grade						
	1	2	3	4	5	N	1	2	3	4	5	N	
Constipation	88	17	3	_	_	108	99	30	_	_	_	129	237
Dental caries	2	2	_	_	_	4	_	_	_	_	_	_	4
Diarrhea	69	26	5	_	_	100	87	22	3	_	_	112	212
Dry mouth	9	_	_	_	_	9	12	1	_	_	_	13	22
Dyspepsia	12	5	_	_	_	17	13	12	_	_	_	25	42
Dysphagia	3	_	2	_	_	5	2	3	_	_	_	5	10
Enterocolitis	1	1	_	_	_	2	_	_	1	_	_	1	3
Esophageal obstruction	_	_	1	_	_	1	_	_	1	_	_	1	2
Esophageal pain	_	_	_	_	_	_	1	3	_	_	_	4	4
Esophageal stenosis	_	1	_	_	_	1	_	_	_	_	_	_	1
Esophageal ulcer	_	_	_	_	_	_	1	_	1	_	_	2	2
Esophagitis	_	_	_	_	_	_	_	1	_	_	_	1	1
Fecal incontinence	3	_	_	_	_	3	4	_	_	_	_	4	7
Flatulence	7	_	_	_	_	7	12	_	_	_	_	12	19
Gastric hemorrhage	_	1	1	_	1	3	_	_	_	_	_	_	3
Gastric ulcer	_	_	1	_	_	1	2	1	_	_	_	3	4
Gastritis	_	1	1	_	_	2	1	2	_	_	_	3	5
Gastroesophageal reflux disease	12	5	_	_	_	17	8	7	_	_	_	15	32
Gastrointestinal disorders - Other, specify	56	17	_	_	_	73	65	30	1	1	_	97	170

System Organ Class/CTCAE Term	Anti-androgen						Enzalutamide						TOTAL
	Worst grade						Worst grade						
	1	2	3	4	5	N	1	2	3	4	5	N	
Gastrointestinal pain	2	—	—	—	—	2	—	3	—	—	—	3	5
Gingival pain	—	—	—	—	—	—	1	—	—	—	—	1	1
Hemorrhoidal hemorrhage	3	—	1	—	—	4	—	—	—	—	—	—	4
Hemorrhoids	6	3	—	—	—	9	3	2	—	—	—	5	14
Lower gastrointestinal hemorrhage	—	—	1	—	—	1	—	—	1	—	—	1	2
Mucositis oral	17	3	2	—	—	22	25	7	2	—	—	34	56
Nausea	67	16	—	—	—	83	114	20	2	—	—	136	219
Oral dysesthesia	1	—	—	—	—	1	2	—	—	—	—	2	3
Oral pain	5	—	—	—	—	5	1	—	—	—	—	1	6
Periodontal disease	1	—	—	—	—	1	—	1	—	—	—	1	2
Proctitis	—	1	—	—	—	1	—	—	—	—	—	—	1
Rectal hemorrhage	6	1	1	—	—	8	17	1	1	—	—	19	27
Rectal mucositis	—	1	—	—	—	1	—	—	—	—	—	—	1
Rectal pain	2	—	—	—	—	2	2	—	—	—	—	2	4
Salivary duct inflammation	4	1	—	—	—	5	3	—	—	—	—	3	8
Small intestinal obstruction	—	—	2	—	—	2	—	—	1	—	—	1	3
Stomach pain	2	1	—	—	—	3	3	—	—	—	—	3	6
Toothache	4	2	—	—	—	6	2	1	—	—	—	3	9
Vomiting	21	10	—	—	—	31	19	7	—	—	—	26	57

System Organ Class/CTCAE Term		Anti-androgen						Enzalutamide						TOTAL
		Worst grade						Worst grade						
		1	2	3	4	5	N	1	2	3	4	5	N	
General disorders and administration site conditions	Total**	592	178	16	_	2	788	654	263	47	_	2	966	1754
	Chills	4	1	_	_	_	5	7	1	_	_	_	8	13
	Death NOS	_	_	_	_	_	_	_	_	_	_	1	1	1
	Edema face	1	_	_	_	_	1	2	_	_	_	_	2	3
	Edema limbs	81	14	_	_	_	95	76	17	_	_	_	93	188
	Edema trunk	_	_	1	_	_	1	3	1	_	_	_	4	5
	Facial pain	_	_	_	_	_	_	1	_	_	_	_	1	1
	Fatigue	279	80	4	_	_	363	292	142	31	_	_	465	828
	Fever	19	5	1	_	_	25	12	4	1	_	_	17	42
	Flu like symptoms	23	10	_	_	_	33	27	6	1	_	_	34	67
	Gait disturbance	3	_	1	_	_	4	9	3	1	_	_	13	17
	General disorders and administration site conditions - Other, specify	89	19	3	_	1	112	123	35	2	_	_	160	272
	Infusion related reaction	2	10	_	_	_	12	1	3	_	_	_	4	16
	Infusion site extravasation	_	1	_	_	_	1	_	3	_	_	_	3	4
	Injection site reaction	6	2	_	_	_	8	4	2	_	_	_	6	14
	Irritability	5	1	_	_	_	6	5	2	_	_	_	7	13
	Localized edema	2	2	1	_	_	5	8	_	_	_	_	8	13
	Malaise	4	1	_	_	_	5	2	_	_	_	_	2	7

System Organ Class/CTCAE Term		Anti-androgen						Enzalutamide						TOTAL
		Worst grade						Worst grade						
		1	2	3	4	5	N	1	2	3	4	5	N	
Hepatobiliary disorders	Neck edema	1	—	—	—	—	1	—	—	—	—	—	—	1
	Non-cardiac chest pain	5	2	—	—	—	7	6	2	1	—	—	9	16
	Pain	68	30	5	—	—	103	76	42	10	—	—	128	231
	Sudden death NOS	—	—	—	—	1	1	—	—	—	—	1	1	2
	Total**	3	2	3	1	—	9	1	2	3	—	—	6	15
	Cholecystitis	—	1	2	1	—	4	—	1	3	—	—	4	8
	Hepatic hemorrhage	—	—	1	—	—	1	—	—	—	—	—	—	1
	Hepatobiliary disorders - Other, specify	3	1	—	—	—	4	1	—	—	—	—	1	5
Immune system disorders	Portal hypertension	—	—	—	—	—	—	—	1	—	—	—	1	1
	Total**	7	3	3	—	—	13	5	11	3	—	—	19	32
	Allergic reaction	5	—	1	—	—	6	2	10	2	—	—	14	20
	Anaphylaxis	—	—	2	—	—	2	—	—	1	—	—	1	3
	Immune system disorders - Other, specify	2	3	—	—	—	5	3	1	—	—	—	4	9
Infections and infestations	Total**	38	172	36	9	3	258	50	205	39	8	—	302	560
	Anorectal infection	—	1	—	—	—	1	—	—	2	—	—	2	3
	Appendicitis	—	—	1	—	—	1	—	—	—	—	—	—	1
	Bladder infection	—	8	—	—	—	8	—	3	—	—	—	3	11
	Bronchial infection	—	8	1	—	—	9	—	8	1	—	—	9	18
	Catheter related infection	—	1	—	—	—	1	—	—	1	1	—	2	3

System Organ Class/CTCAE Term	Anti-androgen						Enzalutamide						TOTAL
	Worst grade						Worst grade						
	1	2	3	4	5	N	1	2	3	4	5	N	
Conjunctivitis infective	—	1	—	—	—	1	—	—	—	—	—	—	1
Corneal infection	—	1	—	—	—	1	—	—	—	—	—	—	1
Endocarditis infective	—	—	—	—	—	—	—	—	1	—	—	1	1
Enterocolitis infectious	—	—	1	—	—	1	—	—	2	—	—	2	3
Esophageal infection	—	—	—	—	—	—	—	1	—	—	—	1	1
Eye infection	—	3	—	—	—	3	—	4	—	—	—	4	7
Gallbladder infection	—	—	—	—	—	—	—	—	1	—	—	1	1
Gum infection	—	—	—	—	—	—	—	4	2	—	—	6	6
Infections and infestations - Other, specify	16	33	2	2	—	53	32	45	3	—	—	80	133
Joint infection	—	—	1	—	—	1	—	1	—	—	—	1	2
Kidney infection	—	—	—	—	—	—	—	—	1	—	—	1	1
Laryngitis	—	—	—	—	—	—	—	2	—	—	—	2	2
Lip infection	1	—	—	—	—	1	—	—	—	—	—	—	1
Lung infection	—	9	8	1	—	18	—	8	7	2	—	17	35
Mucosal infection	6	2	1	—	—	9	5	2	—	—	—	7	16
Nail infection	5	1	—	—	—	6	3	7	—	—	—	10	16
Otitis externa	—	1	—	—	—	1	—	1	—	—	—	1	2
Otitis media	—	2	—	—	—	2	—	2	—	—	—	2	4
Paronychia	1	—	1	—	—	2	2	1	1	—	—	4	6

System Organ Class/CTCAE Term	Anti-androgen						Enzalutamide						TOTAL	
	Worst grade						Worst grade							
	1	2	3	4	5	N	1	2	3	4	5	N		
Injury, poisoning and procedural complications	Penile infection	—	1	—	—	—	1	—	1	—	—	—	1	2
	Pharyngitis	—	—	1	—	—	1	—	1	—	—	—	1	2
	Rash pustular	—	1	—	—	—	1	—	2	—	—	—	2	3
	Rhinitis infective	—	2	—	—	—	2	—	1	—	—	—	1	3
	Salivary gland infection	—	—	—	—	—	—	—	1	—	—	—	1	1
	Scrotal infection	—	2	—	—	—	2	—	5	—	—	—	5	7
	Sepsis	—	—	1	6	2	9	—	—	—	5	—	5	14
	Sinusitis	—	9	—	—	—	9	—	10	—	—	—	10	19
	Skin infection	7	13	8	—	—	28	7	12	6	—	—	25	53
	Soft tissue infection	—	—	1	—	—	1	—	—	1	—	—	1	2
	Tooth infection	—	5	1	—	—	6	—	9	—	—	—	9	15
	Upper respiratory infection	2	43	2	—	—	47	1	48	3	—	—	52	99
	Urinary tract infection	—	21	6	—	1	28	—	22	7	—	—	29	57
	Wound infection	—	4	—	—	—	4	—	4	—	—	—	4	8
	Total**	24	26	9	—	—	59	48	63	25	3	—	139	198
Ankle fracture	1	—	—	—	—	1	1	2	—	—	—	3	4	
Bruising	4	—	—	—	—	4	6	2	—	—	—	8	12	
Burn	—	—	—	—	—	—	—	2	1	—	—	3	3	
Dermatitis radiation	—	—	—	—	—	—	1	—	—	—	—	1	1	

System Organ Class/CTCAE Term		Anti-androgen						Enzalutamide						TOTAL
		Worst grade						Worst grade						
		1	2	3	4	5	N	1	2	3	4	5	N	
Investigations	Fall	10	8	2	_	_	20	20	28	6	_	_	54	74
	Fracture	2	5	3	_	_	10	2	11	8	1	_	22	32
	Hip fracture	_	_	1	_	_	1	_	1	2	_	_	3	4
	Injury, poisoning and procedural complications - Other, specify	7	11	2	_	_	20	14	11	3	_	_	28	48
	Intestinal stoma site bleeding	_	_	_	_	_	_	_	1	_	_	_	1	1
	Intraoperative gastrointestinal injury	_	_	_	_	_	_	_	_	_	1	_	1	1
	Postoperative hemorrhage	_	_	_	_	_	_	_	1	_	_	_	1	1
	Spinal fracture	_	_	_	_	_	_	_	3	4	1	_	8	8
	Wound complication	_	_	_	_	_	_	_	_	1	_	_	1	1
	Wound dehiscence	_	1	_	_	_	1	_	_	_	_	_	_	1
	Wrist fracture	_	1	1	_	_	2	4	1	_	_	_	5	7
	Total**	241	57	18	14	_	330	221	63	34	14	_	332	662
	Alanine aminotransferase increased	35	7	3	_	_	45	20	3	3	_	_	26	71
	Alkaline phosphatase increased	29	3	1	_	_	33	26	7	4	_	_	37	70
	Aspartate aminotransferase increased	31	5	1	_	_	37	8	2	1	_	_	11	48
Blood bilirubin increased	3	_	_	1	_	4	1	2	_	_	_	3	7	
Cholesterol high	28	1	_	1	_	30	31	5	_	_	_	36	66	
Creatinine increased	22	3	_	_	_	25	29	2	_	_	_	31	56	

System Organ Class/CTCAE Term	Anti-androgen						Enzalutamide						TOTAL	
	Worst grade						Worst grade							
	1	2	3	4	5	N	1	2	3	4	5	N		
Metabolism and nutrition disorders	Electrocardiogram QT corrected interval prolonged	1	—	—	—	—	1	—	—	—	—	—	—	1
	Forced expiratory volume decreased	—	—	—	—	—	—	1	—	—	—	—	1	1
	GGT increased	11	5	3	1	—	20	9	2	3	—	—	14	34
	Hemoglobin increased	—	—	—	—	—	—	2	—	—	—	—	2	2
	Investigations - Other, specify	21	12	—	—	—	33	23	11	—	—	—	34	67
	Lymphocyte count decreased	7	1	—	—	—	8	8	1	—	—	—	9	17
	Lymphocyte count increased	—	1	—	—	—	1	—	—	—	—	—	—	1
	Neutrophil count decreased	10	4	6	10	—	30	13	5	17	14	—	49	79
	Pancreatic enzymes decreased	—	—	—	—	—	—	—	—	1	—	—	1	1
	Platelet count decreased	5	1	—	—	—	6	6	—	1	—	—	7	13
	Weight gain	26	9	3	—	—	38	24	10	2	—	—	36	74
	Weight loss	10	1	—	—	—	11	15	9	—	—	—	24	35
	White blood cell decreased	2	4	1	1	—	8	5	4	2	—	—	11	19
	Total**	122	46	11	3	—	182	146	53	15	2	1	217	399
	Acidosis	—	—	—	—	—	—	—	—	—	—	1	1	1
	Alcohol intolerance	—	—	—	—	—	—	—	1	—	—	—	1	1
	Anorexia	31	12	—	—	—	43	60	20	1	—	—	81	124
Dehydration	1	2	2	—	—	5	3	1	1	—	—	5	10	
Glucose intolerance	—	1	—	—	—	1	—	—	—	—	—	—	1	

System Organ Class/CTCAE Term		Anti-androgen						Enzalutamide						TOTAL
		Worst grade						Worst grade						
		1	2	3	4	5	N	1	2	3	4	5	N	
Metabolism and nutrition disorders	Hypercalcemia	8	_	1	_	_	9	13	1	1	_	_	15	24
	Hyperglycemia	10	10	4	2	_	26	12	4	6	_	_	22	48
	Hyperkalemia	2	3	2	_	_	7	9	1	_	_	_	10	17
	Hypermagnesemia	2	_	_	_	_	2	_	_	_	_	_	_	2
	Hypernatremia	1	_	_	_	_	1	5	_	1	_	_	6	7
	Hypertriglyceridemia	12	1	1	1	_	15	10	5	1	_	_	16	31
	Hyperuricemia	1	_	_	_	_	1	1	_	_	_	_	1	2
	Hypoalbuminemia	1	2	_	_	_	3	1	1	_	_	_	2	5
	Hypocalcemia	5	_	_	_	_	5	2	2	1	_	_	5	10
	Hypoglycemia	5	1	_	_	_	6	1	1	_	_	_	2	8
	Hypokalemia	4	5	1	_	_	10	6	_	_	_	_	6	16
	Hypomagnesemia	4	_	_	_	_	4	1	2	1	_	_	4	8
	Hyponatremia	3	_	_	_	_	3	3	_	1	2	_	6	9
	Hypophosphatemia	2	_	_	_	_	2	_	1	_	_	_	1	3
	Metabolism and nutrition disorders - Other, specify	30	7	_	_	_	37	19	11	_	_	_	30	67
	Obesity	_	2	_	_	_	2	_	2	1	_	_	3	5
	Musculoskeletal and connective tissue disorders	Total**	502	224	35	_	_	761	579	236	42	_	_	857
Arthralgia		63	9	2	_	_	74	62	18	3	_	_	83	157
Arthritis		23	10	9	_	_	42	24	4	8	_	_	36	78

System Organ Class/CTCAE Term	Anti-androgen						Enzalutamide						TOTAL
	Worst grade						Worst grade						
	1	2	3	4	5	N	1	2	3	4	5	N	
Avascular necrosis	—	1	—	—	—	1	—	1	—	—	—	1	2
Back pain	89	50	10	—	—	149	103	56	11	—	—	170	319
Bone pain	32	22	2	—	—	56	35	12	3	—	—	50	106
Buttock pain	5	3	1	—	—	9	6	1	—	—	—	7	16
Chest wall pain	12	3	—	—	—	15	9	4	—	—	—	13	28
Flank pain	5	5	1	—	—	11	7	2	—	—	—	9	20
Generalized muscle weakness	13	4	—	—	—	17	26	2	2	—	—	30	47
Joint effusion	2	—	—	—	—	2	1	1	—	—	—	2	4
Joint range of motion decreased	3	1	—	—	—	4	5	—	—	—	—	5	9
Muscle weakness left-sided	—	—	—	—	—	—	—	1	—	—	—	1	1
Muscle weakness lower limb	10	4	—	—	—	14	20	3	2	—	—	25	39
Muscle weakness right-sided	1	—	—	—	—	1	—	1	—	—	—	1	2
Muscle weakness trunk	1	—	—	—	—	1	—	—	—	—	—	—	1
Muscle weakness upper limb	4	1	—	—	—	5	5	4	—	—	—	9	14
Musculoskeletal and connective tissue disorder - Other, specify	140	72	6	—	—	218	148	79	7	—	—	234	452
Musculoskeletal deformity	1	—	—	—	—	1	—	—	1	—	—	1	2
Myalgia	41	11	—	—	—	52	49	11	2	—	—	62	114
Neck pain	11	1	1	—	—	13	27	10	1	—	—	38	51

System Organ Class/CTCAE Term		Anti-androgen						Enzalutamide						TOTAL
		Worst grade						Worst grade						
		1	2	3	4	5	N	1	2	3	4	5	N	
Neoplasms benign, malignant and unspecified (incl cysts and polyps)	Osteonecrosis of jaw	2	3	_	_	_	5	_	2	_	_	_	2	7
	Osteoporosis	2	3	_	_	_	5	6	6	_	_	_	12	17
	Pain in extremity	42	21	3	_	_	66	45	17	2	_	_	64	130
	Scoliosis	_	_	_	_	_	_	1	_	_	_	_	1	1
	Soft tissue necrosis lower limb	_	_	_	_	_	_	_	1	_	_	_	1	1
	Total**	7	6	7	4	_	24	7	18	11	1	_	37	61
	Neoplasms benign, malignant and unspecified (incl cysts and polyps) - Other, spe	7	6	7	4	_	24	6	17	9	1	_	33	57
Nervous system disorders	Tumor pain	_	_	_	_	_	_	1	1	2	_	_	4	4
	Total**	295	71	26	_	_	392	535	138	46	1	1	721	1113
	Akathisia	1	_	_	_	_	1	2	_	_	_	_	2	3
	Amnesia	1	2	_	_	_	3	7	4	_	_	_	11	14
	Ataxia	2	1	2	_	_	5	3	2	_	_	_	5	10
	Cognitive disturbance	2	1	_	_	_	3	10	3	_	_	_	13	16
	Concentration impairment	6	_	_	_	_	6	21	3	_	_	_	24	30
	Dizziness	36	5	_	_	_	41	66	11	1	_	_	78	119
	Dysarthria	_	2	_	_	_	2	1	_	_	_	_	1	3
	Dysesthesia	1	_	_	_	_	1	2	_	_	_	_	2	3
	Dysgeusia	19	4	_	_	_	23	44	4	_	_	_	48	71

System Organ Class/CTCAE Term	Anti-androgen						Enzalutamide						TOTAL
	Worst grade						Worst grade						
	1	2	3	4	5	N	1	2	3	4	5	N	
Dysphasia	—	—	—	—	—	—	1	1	—	—	—	2	2
Extrapyramidal disorder	—	—	—	—	—	—	2	—	—	—	—	2	2
Facial muscle weakness	2	—	—	—	—	2	—	—	—	—	—	—	2
Facial nerve disorder	1	—	1	—	—	2	—	—	1	—	—	1	3
Headache	42	7	2	—	—	51	61	13	1	—	—	75	126
Hypersomnia	—	—	—	—	—	—	—	1	—	—	—	1	1
Intracranial hemorrhage	—	1	—	—	—	1	—	—	2	1	—	3	4
Ischemia cerebrovascular	1	—	—	—	—	1	—	—	1	—	—	1	2
Lethargy	7	2	—	—	—	9	8	4	—	—	—	12	21
Memory impairment	19	1	—	—	—	20	50	12	—	—	—	62	82
Movements involuntary	1	—	—	—	—	1	4	1	—	—	—	5	6
Nervous system disorders - Other, specify	28	11	1	—	—	40	61	11	5	—	—	77	117
Neuralgia	4	3	—	—	—	7	4	1	—	—	—	5	12
Oculomotor nerve disorder	—	—	1	—	—	1	—	—	—	—	—	—	1
Paresthesia	19	7	—	—	—	26	37	3	1	—	—	41	67
Peripheral motor neuropathy	2	1	—	—	—	3	13	7	1	—	—	21	24
Peripheral sensory neuropathy	98	17	1	—	—	116	115	37	4	—	—	156	272
Presyncope	—	2	—	—	—	2	—	9	—	—	—	9	11
Pyramidal tract syndrome	—	—	6	—	—	6	—	1	3	—	—	4	10

System Organ Class/CTCAE Term	Anti-androgen						Enzalutamide						TOTAL	
	Worst grade						Worst grade							
	1	2	3	4	5	N	1	2	3	4	5	N		
Neurologic disorders	Radiculitis	—	1	—	—	—	1	1	1	1	—	—	3	4
	Seizure	—	—	—	—	—	—	1	4	2	—	—	7	7
	Sinus pain	—	—	—	—	—	—	2	—	—	—	—	2	2
	Somnolence	—	—	—	—	—	—	2	—	—	—	—	2	2
	Spasticity	—	—	—	—	—	—	—	1	—	—	—	1	1
	Stroke	—	—	2	—	—	2	1	2	—	—	1	4	6
	Syncope	1	—	6	—	—	7	—	—	20	—	—	20	27
	Transient ischemic attacks	1	2	—	—	—	3	4	1	—	—	—	5	8
	Tremor	1	1	—	—	—	2	12	1	—	—	—	13	15
	Vasovagal reaction	—	—	4	—	—	4	—	—	3	—	—	3	7
Psychiatric disorders	Total**	176	61	3	—	—	240	243	86	6	5	—	340	580
Psychiatric disorders	Agitation	6	1	—	—	—	7	10	—	—	1	—	11	18
	Anxiety	26	6	1	—	—	33	32	14	—	—	—	46	79
	Confusion	7	3	—	—	—	10	6	6	—	—	—	12	22
	Delirium	2	2	1	—	—	5	1	—	—	—	—	1	6
	Depression	33	10	—	—	—	43	37	17	2	2	—	58	101
	Hallucinations	—	—	—	—	—	—	1	—	—	—	—	1	1
	Insomnia	57	29	1	—	—	87	91	22	4	—	—	117	204
	Libido decreased	13	2	—	—	—	15	14	7	—	—	—	21	36

System Organ Class/CTCAE Term	Anti-androgen						Enzalutamide						TOTAL	
	Worst grade						Worst grade							
	1	2	3	4	5	N	1	2	3	4	5	N		
Renal and urinary disorders	Libido increased	—	—	—	—	—	—	1	—	—	—	—	1	1
	Mania	—	—	—	—	—	—	—	—	—	1	—	1	1
	Personality change	2	—	—	—	—	2	1	—	—	—	—	1	3
	Psychiatric disorders - Other, specify	28	8	—	—	—	36	46	13	—	—	—	59	95
	Psychosis	—	—	—	—	—	—	1	—	—	—	—	1	1
	Restlessness	2	—	—	—	—	2	2	5	—	—	—	7	9
	Suicidal ideation	—	—	—	—	—	—	—	2	—	—	—	2	2
	Suicide attempt	—	—	—	—	—	—	—	—	—	1	—	1	1
	Total**	231	67	22	1	—	321	258	91	38	—	—	387	708
	Acute kidney injury	3	1	1	—	—	5	1	—	5	—	—	6	11
	Bladder spasm	1	2	—	—	—	3	2	1	—	—	—	3	6
	Chronic kidney disease	2	—	—	—	—	2	—	—	—	—	—	—	2
	Cystitis noninfective	6	—	—	—	—	6	5	2	2	—	—	9	15
	Hematuria	20	7	5	—	—	32	31	8	8	—	—	47	79
	Renal and urinary disorders - Other, specify	59	11	1	—	—	71	67	25	1	—	—	93	164
Renal calculi	2	2	2	1	—	7	2	3	4	—	—	9	16	
Renal colic	—	—	1	—	—	1	—	—	—	—	—	—	1	
Urinary fistula	—	—	—	—	—	—	—	—	1	—	—	1	1	
Urinary frequency	86	16	—	—	—	102	94	26	—	—	—	120	222	

System Organ Class/CTCAE Term		Anti-androgen						Enzalutamide						TOTAL
		Worst grade						Worst grade						
		1	2	3	4	5	N	1	2	3	4	5	N	
Reproductive system and breast disorders	Urinary incontinence	25	6	2	_	_	33	24	12	2	_	_	38	71
	Urinary retention	6	14	5	_	_	25	4	5	6	_	_	15	40
	Urinary tract obstruction	1	3	5	_	_	9	1	3	8	_	_	12	21
	Urinary tract pain	5	2	_	_	_	7	3	1	1	_	_	5	12
	Urinary urgency	13	3	_	_	_	16	24	5	_	_	_	29	45
	Urine discoloration	2	_	_	_	_	2	_	_	_	_	_	_	2
	Total**	82	27	12	_	_	121	119	34	13	_	_	166	287
	Breast atrophy	1	_	_	_	_	1	_	_	_	_	_	_	1
	Breast pain	5	1	_	_	_	6	6	_	_	_	_	6	12
	Erectile dysfunction	14	17	12	_	_	43	13	19	12	_	_	44	87
	Gynecomastia	24	5	_	_	_	29	30	10	_	_	_	40	69
	Pelvic pain	1	_	_	_	_	1	4	_	_	_	_	4	5
	Penile pain	3	_	_	_	_	3	3	_	_	_	_	3	6
	Perineal pain	1	_	_	_	_	1	1	_	_	_	_	1	2
	Prostatic obstruction	_	_	_	_	_	_	_	_	1	_	_	1	1
Prostatic pain	1	_	_	_	_	1	_	_	_	_	_	_	1	
Reproductive system and breast disorders - Other, specify	23	4	_	_	_	27	32	2	_	_	_	34	61	
Scrotal pain	1	_	_	_	_	1	3	_	_	_	_	3	4	

System Organ Class/CTCAE Term		Anti-androgen						Enzalutamide						TOTAL
		Worst grade						Worst grade						
		1	2	3	4	5	N	1	2	3	4	5	N	
Respiratory, thoracic and mediastinal disorders	Testicular disorder	1	—	—	—	—	1	3	—	—	—	—	3	4
	Testicular pain	7	—	—	—	—	7	24	3	—	—	—	27	34
	Total**	192	55	5	1	—	253	297	73	7	—	1	378	631
	Allergic rhinitis	4	1	—	—	—	5	6	1	—	—	—	7	12
	Atelectasis	—	—	—	—	—	—	1	—	—	—	—	1	1
	Bronchopulmonary hemorrhage	2	—	—	—	—	2	—	—	—	—	—	—	2
	Bronchospasm	—	1	—	—	—	1	—	—	—	—	—	—	1
	Cough	61	10	—	—	—	71	70	8	—	—	—	78	149
	Dyspnea	50	9	1	1	—	61	74	26	2	—	—	102	163
	Epistaxis	12	4	—	—	—	16	22	1	—	—	—	23	39
	Hiccups	5	2	—	—	—	7	—	—	—	—	—	—	7
	Hoarseness	1	—	—	—	—	1	8	1	—	—	—	9	10
	Hypoxia	—	1	1	—	—	2	—	—	—	—	—	—	2
	Laryngeal fistula	1	—	—	—	—	1	—	—	—	—	—	—	1
	Laryngeal mucositis	—	—	—	—	—	—	1	—	—	—	—	1	1
	Laryngopharyngeal dysesthesia	—	—	—	—	—	—	1	—	—	—	—	1	1
	Nasal congestion	5	—	—	—	—	5	7	1	—	—	—	8	13
	Pleural effusion	2	1	—	—	—	3	—	2	1	—	—	3	6
	Pleuritic pain	—	—	—	—	—	—	1	—	—	—	—	1	1

System Organ Class/CTCAE Term	Anti-androgen						Enzalutamide						TOTAL	
	Worst grade						Worst grade							
	1	2	3	4	5	N	1	2	3	4	5	N		
	Pneumonitis	—	1	1	—	—	2	—	3	1	—	—	4	6
	Pneumothorax	—	—	—	—	—	—	1	—	—	—	—	1	1
	Postnasal drip	2	—	—	—	—	2	8	2	—	—	—	10	12
	Productive cough	2	2	—	—	—	4	3	4	—	—	—	7	11
	Pulmonary edema	—	—	—	—	—	—	2	—	1	—	—	3	3
	Pulmonary fibrosis	—	—	—	—	—	—	1	—	—	—	—	1	1
	Pulmonary hypertension	1	—	—	—	—	1	—	—	—	—	—	—	1
	Respiratory, thoracic and mediastinal disorders - Other, specify	31	16	1	—	—	48	68	20	1	—	1	90	138
	Retinoic acid syndrome	—	—	—	—	—	—	1	—	—	—	—	1	1
	Sinus disorder	2	1	—	—	—	3	2	1	—	—	—	3	6
	Sleep apnea	1	2	1	—	—	4	1	2	1	—	—	4	8
	Sneezing	—	—	—	—	—	—	4	—	—	—	—	4	4
	Sore throat	5	4	—	—	—	9	8	2	—	—	—	10	19
	Voice alteration	1	—	—	—	—	1	1	—	—	—	—	1	2
	Wheezing	4	—	—	—	—	4	6	—	—	—	—	6	10
Skin and subcutaneous tissue disorders	Total**	269	63	5	—	—	337	324	103	6	—	—	433	770
	Alopecia	52	14	—	—	—	66	63	24	—	—	—	87	153
	Bullous dermatitis	—	1	—	—	—	1	—	—	—	—	—	—	1

System Organ Class/CTCAE Term	Anti-androgen						Enzalutamide						TOTAL
	Worst grade						Worst grade						
	1	2	3	4	5	N	1	2	3	4	5	N	
Dry skin	31	3	_	_	_	34	41	5	_	_	_	46	80
Erythema multiforme	3	1	_	_	_	4	3	1	_	_	_	4	8
Erythroderma	_	1	_	_	_	1	_	1	_	_	_	1	2
Hyperhidrosis	8	_	_	_	_	8	2	_	_	_	_	2	10
Nail discoloration	16	_	_	_	_	16	30	_	_	_	_	30	46
Nail loss	8	_	_	_	_	8	5	2	_	_	_	7	15
Nail ridging	17	_	_	_	_	17	15	_	_	_	_	15	32
Pain of skin	1	_	_	_	_	1	3	_	_	_	_	3	4
Palmar-plantar erythrodysesthesia syndrome	5	2	1	_	_	8	8	6	_	_	_	14	22
Periorbital edema	_	_	_	_	_	_	1	_	_	_	_	1	1
Photosensitivity	2	_	_	_	_	2	2	1	_	_	_	3	5
Pruritus	12	4	_	_	_	16	23	2	_	_	_	25	41
Purpura	1	1	_	_	_	2	_	_	_	_	_	_	2
Rash acneiform	8	1	_	_	_	9	4	1	_	_	_	5	14
Rash maculo-papular	14	2	1	_	_	17	26	5	3	_	_	34	51
Scalp pain	1	_	_	_	_	1	_	_	_	_	_	_	1
Skin and subcutaneous tissue disorders - Other, specify	84	30	2	_	_	116	86	49	1	_	_	136	252
Skin atrophy	_	_	_	_	_	_	2	_	_	_	_	2	2

System Organ Class/CTCAE Term		Anti-androgen						Enzalutamide							TOTAL
		Worst grade						Worst grade							
		1	2	3	4	5	N	1	2	3	4	5	N		
Social circumstances	Skin hyperpigmentation	2	—	—	—	—	2	2	—	—	—	—	2	4	
	Skin hypopigmentation	1	—	—	—	—	1	1	1	—	—	—	2	3	
	Skin ulceration	2	1	1	—	—	4	4	3	1	—	—	8	12	
	Urticaria	1	2	—	—	—	3	3	2	1	—	—	6	9	
	Total**	4	—	—	—	—	4	2	—	—	—	—	2	6	
	Social circumstances - Other, specify	4	—	—	—	—	4	2	—	—	—	—	2	6	
Surgical and medical procedures	Total**	4	21	10	—	—	35	9	27	13	—	—	49	84	
	Surgical and medical procedures - Other, specify	4	21	10	—	—	35	9	27	13	—	—	49	84	
Vascular disorders	Total**	377	120	34	1	—	532	388	172	59	—	—	619	1151	
	Flushing	34	6	—	—	—	40	37	4	—	—	—	41	81	
	Hematoma	3	—	—	—	—	3	3	—	—	—	—	3	6	
	Hot flashes	281	63	1	—	—	345	295	83	4	—	—	382	727	
	Hypertension	15	30	24	1	—	70	15	60	43	—	—	118	188	
	Hypotension	7	5	2	—	—	14	5	6	3	—	—	14	28	
	Lymphedema	7	2	1	—	—	10	3	—	—	—	—	3	13	
	Peripheral ischemia	—	2	1	—	—	3	—	—	1	—	—	1	4	
	Phlebitis	—	2	—	—	—	2	—	1	—	—	—	1	3	
	Superficial thrombophlebitis	—	1	—	—	—	1	—	1	—	—	—	1	2	
	Superior vena cava syndrome	—	—	—	—	—	—	—	1	—	—	—	1	1	

System Organ Class/CTCAE Term	Anti-androgen						Enzalutamide						TOTAL
	Worst grade						Worst grade						
	1	2	3	4	5	N	1	2	3	4	5	N	
Thromboembolic event	1	3	4	_	_	8	1	4	4	_	_	9	17
Vascular disorders - Other, specify	29	6	1	_	_	36	29	12	2	_	_	43	79
Visceral arterial ischemia	_	_	_	_	_	_	_	_	2	_	_	2	2

* *If a patient had multiple events within a particular term, that with the worst grade is shown.*

** *The worst grade events are summed within a system organ class to form row totals*

Supplementary Table S7. Adverse Events in first six months

Number of adverse events, and number of patients experiencing adverse events, within the first six months of study entry.

Number of AEs by Grade		
Grade	Anti-androgen	Enzalutamide
Grade 1	3972	4795
Grade 2	844	1175
Grade 3	194	247
Grade 4	31	32
Grade 5	3	2
Total	5044	6251

Number of Patients with AEs by Worst Grade		
Grade*	Anti-androgen	Enzalutamide
Grade 1	183	142
Grade 2	199	225
Grade 3	107	151
Grade 4	29	25
Grade 5	3	2
Total	521	545

* If a participant had multiple events, the event with the worst grade is counted.

Supplementary Table S8: Adverse events of interest in docetaxel stratum within 6 months of randomization

		Anti-androgen												Enzalutamide											
		With Docetaxel						Without Docetaxel						With Docetaxel						Without Docetaxel					
	Worst Grade	1	2	3	4	5	N	1	2	3	4	5	N	1	2	3	4	5	N	1	2	3	4	5	N
Blood and lymphatic system disorders	Total**	40	10	25	9	0	84	22	3	1	0	0	26	38	16	28	7	0	89	23	4	2	0	0	29
	Anemia	34	6	1	1	0	42	16	2	1	0	0	19	30	13	1	0	0	44	17	1	1	0	0	19
	Other	6	4	0	0	0	10	6	1	0	0	0	7	7	2	0	0	0	9	6	3	0	0	0	9
	Bone marrow hypocellular	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0	0
	Febrile neutropenia	0	0	24	8	0	32	0	0	0	0	0	0	0	0	28	7	0	35	0	0	1	0	0	1
Ear and labyrinth disorders	Total**	1	0	0	0	0	1	0	0	0	0	0	0	4	1	0	0	0	5	2	3	0	0	0	5
	Hearing impaired	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	2	0	0	0	0	0	0
	Tinnitus	1	0	0	0	0	1	0	0	0	0	0	0	2	1	0	0	0	3	2	3	0	0	0	5
Eye disorders	Total**	13	2	0	0	0	15	0	0	0	0	0	0	48	4	0	0	0	52	1	0	0	0	0	1
	Watering eyes	13	2	0	0	0	15	0	0	0	0	0	0	48	4	0	0	0	52	1	0	0	0	0	1
Gastrointestinal disorders	Total**	88	24	5	0	0	117	28	5	0	0	0	33	103	20	2	0	0	125	54	11	1	0	0	66
	Diarrhea	38	14	3	0	0	55	10	4	0	0	0	14	34	12	0	0	0	46	19	5	0	0	0	24
	Mucositis oral	16	3	2	0	0	21	0	0	0	0	0	0	20	4	2	0	0	26	2	1	0	0	0	3
	Nausea	34	7	0	0	0	41	18	1	0	0	0	19	49	4	0	0	0	53	33	5	1	0	0	39
General disorders and administration site conditions	Total**	180	58	2	0	0	240	115	10	1	0	0	126	186	67	8	0	0	261	135	36	7	0	0	178
	Edema limbs	34	8	0	0	0	42	14	1	0	0	0	15	34	7	0	0	0	41	7	4	0	0	0	11
	Fatigue	130	35	1	0	0	166	99	9	1	0	0	109	140	52	7	0	0	199	128	32	7	0	0	167
	Fever	14	4	1	0	0	19	2	0	0	0	0	2	11	2	1	0	0	14	0	0	0	0	0	0
	Infusion related reaction	2	10	0	0	0	12	NA	NA	NA	NA	NA	NA	1	3	0	0	0	4	NA	NA	NA	NA	NA	NA
	Infusion site extravasation	0	1	0	0	0	1	0	0	0	0	0	0	0	3	0	0	0	3	0	0	0	0	0	0

		Anti-androgen												Enzalutamide											
		With Docetaxel						Without Docetaxel						With Docetaxel						Without Docetaxel					
	Worst Grade	1	2	3	4	5	N	1	2	3	4	5	N	1	2	3	4	5	N	1	2	3	4	5	N
Immune system disorders	Total**	0	0	2	0	0	2	4	0	0	0	0	4	0	5	0	0	0	5	1	0	0	0	0	1
	Allergic reaction	0	0	1	0	0	1	4	0	0	0	0	4	0	5	0	0	0	5	1	0	0	0	0	1
	Anaphylaxis	0	0	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Infections and infestations	Total**	13	43	9	6	1	72	9	24	7	1	1	42	18	31	11	3	0	63	6	31	7	0	0	44
	Anorectal infection	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	2	0	0	0	0	0	0
	Bladder infection	0	2	0	0	0	2	0	4	0	0	0	4	0	0	0	0	0	0	0	1	0	0	0	1
	Bronchial infection	0	3	0	0	0	3	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	2
	Catheter related infection	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	2	0	0	0	0	0	0
	Conjunctivitis infective	0	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Enterocolitis infectious	0	0	0	0	0	0	0	0	1	0	0	1	0	0	1	0	0	1	0	0	0	0	0	0
	Eye infection	0	2	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1
	Gum infection	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	0	0	0	0	0
	Infections and infestations - Other, specify	3	7	1	2	0	13	6	2	0	0	0	8	12	9	0	0	0	21	5	7	1	0	0	13
	Lip infection	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Lung infection	0	2	2	0	0	4	0	1	1	0	0	2	0	1	2	0	0	3	0	1	1	0	0	2
	Mucosal infection	5	2	1	0	0	8	0	0	0	0	0	0	4	0	0	0	0	4	0	1	0	0	0	1
	Nail infection	1	1	0	0	0	2	0	0	0	0	0	0	1	4	0	0	0	5	0	1	0	0	0	1
	Paronychia	1	0	1	0	0	2	0	0	0	0	0	0	1	0	1	0	0	2	0	0	0	0	0	0
	Penile infection	0	0	0	0	0	0	0	1	0	0	0	1	0	1	0	0	0	1	0	0	0	0	0	0
	Pharyngitis	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	0	0	0	0	0	0
	Rash pustular	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	0	0	0	0	0	0
	Rhinitis infective	0	2	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Salivary gland infection	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	0	0	0	0	0	0
	Sepsis	0	0	0	4	1	5	0	0	0	1	0	1	0	0	0	2	0	2	0	0	0	0	0	0
	Sinusitis	0	1	0	0	0	1	0	2	0	0	0	2	0	1	0	0	0	1	0	2	0	0	0	2

		Anti-androgen												Enzalutamide											
		With Docetaxel						Without Docetaxel						With Docetaxel						Without Docetaxel					
	Worst Grade	1	2	3	4	5	N	1	2	3	4	5	N	1	2	3	4	5	N	1	2	3	4	5	N
	Skin infection	2	4	3	0	0	9	3	1	1	0	0	5	0	2	1	0	0	3	1	3	1	0	0	5
	Tooth infection	0	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1
	Upper respiratory infection	0	9	0	0	0	9	0	7	1	0	0	8	0	6	0	0	0	6	0	5	1	0	0	6
	Urinary tract infection	0	6	1	0	0	7	0	6	3	0	1	10	0	4	2	0	0	6	0	6	3	0	0	9
Investigations	Total**	9	6	5	11	0	31	4	0	0	0	0	4	4	8	19	13	0	44	7	0	0	1	0	8
	Neutrophil count decreased	5	3	4	10	0	22	2	0	0	0	0	2	1	4	17	13	0	35	4	0	0	1	0	5
	Platelet count decreased	2	0	0	0	0	2	2	0	0	0	0	2	0	0	0	0	0	0	2	0	0	0	0	2
	White blood cell decreased	2	3	1	1	0	7	0	0	0	0	0	0	3	4	2	0	0	9	1	0	0	0	0	1
Musculoskeletal and connective tissue disorders	Total**	25	7	0	0	0	32	5	2	0	0	0	7	21	3	1	0	0	25	14	4	0	0	0	18
	Myalgia	25	7	0	0	0	32	5	2	0	0	0	7	21	3	1	0	0	25	14	4	0	0	0	18
Nervous system disorders	Total**	69	9	1	0	0	79	16	2	0	0	0	18	99	29	3	0	0	131	20	0	1	0	0	21
	Dysesthesia	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1
	Paresthesia	5	1	0	0	0	6	6	0	0	0	0	6	13	1	0	0	0	14	9	0	0	0	0	9
	Peripheral motor neuropathy	1	1	0	0	0	2	0	0	0	0	0	0	9	4	0	0	0	13	0	0	1	0	0	1
	Peripheral sensory neuropathy	62	7	1	0	0	70	10	2	0	0	0	12	77	24	3	0	0	104	10	0	0	0	0	10
Skin and subcutaneous tissue disorders	Total**	61	13	0	0	0	74	7	0	0	0	0	7	79	21	0	0	0	100	3	2	0	0	0	5
	Alopecia	44	13	0	0	0	57	5	0	0	0	0	5	50	21	0	0	0	71	3	1	0	0	0	4
	Nail discoloration	13	0	0	0	0	13	0	0	0	0	0	0	25	0	0	0	0	25	0	0	0	0	0	0
	Nail loss	4	0	0	0	0	4	2	0	0	0	0	2	4	0	0	0	0	4	0	1	0	0	0	1

*If a patient had multiple events within a particular term, that with the worst grade is shown.

**The worst grade events are summed within a system organ class to form row totals

Abbreviation: NA, not applicable.